Leonid Kondrashov

About

Leonid is a Ph.D. candidate at Nanyang Technological University with two years of internship experience at Intel, specializing in serverless systems and performance analysis. Proficient in Go, C++, and Python, with proven skills in cloud computing and computer architecture.

Personal info

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Education

- Ph.D. (01.2024-now) at Nanyang Technological University, Singapore
  - Researching serverless system optimizations: autoscaling, snapshot state management
  - **PulseNet**: an efficient cluster manager that achieves high scaling performance for FaaS, together with full-feature support of conventional cluster managers
- B.Sc. (with Honours), Applied Math and Physics (09.2018-07.2022) at MIPT (Moscow Institute of Physics and Technology)
  - GPA: 4.95/5.00
  - B.Sc. thesis: “Methods of analysis and comparison of execution speed of cycle-accurate simulators of microprocessor”. Advisor: Pavel Kryukov (Intel/MIPT).
  - Obtained skills: programming, electrical engineering, OS, computer networks

Publications

- **Kondrashov, L.**, Liu, H., Park, J., Zhou, B., Liu, Z., Lu, C., Mancini, R., Choukse, E., Javaid, H., Sviridov, G., Peng, T., Zhao, C., Avdeeva, A., Gusev, A., Kogias, M., Mai, L., and Ustiugov, D., 2026. Rethinking AI Cloud Infrastructure for Agentic Serving Systems with the Aries Experimentation Framework. SoCC’26, https://doi.org/10.1145/3828161.3857887
- **Kondrashov, L.**, Zhang, L., Jesus, AFP., Pistor, H., Cali, M., Mancini, R., Bruno, R., and Ustiugov, D., 2026. SnapShare: Isolation-Preserving Deduplication for Serverless Snapshots. arXiv preprint arXiv:2610.13318
- **Kondrashov, L.**, Zhou, B., Wang, H., and Ustiugov, D., 2025. The High Cost of Keeping Warm: Characterizing Overhead in Serverless Autoscaling Policies. arXiv preprint arXiv:2509.03104
- **Kondrashov, L.**, Cvetković, L., Wang, H., Zhou, B., Rungta, D., and Ustiugov, D., 2025. Melding the Serverless Control Plane with the Conventional Cluster Manager for Speed and Compatibility. arXiv preprint arXiv:2505.24551

Workshops

- Kondrashov, L., and Ustiugov, D., 2024. Quantifying the System Tax in Serverless Cloud. SESAME @ EuroSys'24

Work experience

- Intel Software Intern (07.2020-07.2022) in Core Microarchitecture Team
    - Analyzed the performance of perspective microarchitectural features in cache prefetching and cache structures for Intel Core microarchitecture
    - Enhanced the speed and accuracy of the microprocessor simulator

Teaching experience

- Mentoring for Technotrack course “Advanced C++” (02.2020-05.2020)
    - Checking students’ assignments
    - Answering students’ questions about lecture material
- Mentoring for STM32 programming course (09.2019–01.2020)
    - Teaching students the basics of STM32 microcontroller programming based on pre-recorded lectures
    - Writing descriptions for several example lab works

Projects

- ARIES agentic experimentation platform
    - Maintenance and release management
    - Code: github.com/hyscale-lab/ARIES
- vHive serverless platform maintenance
    - Performance and reliability enhancements
    - Onboarding of new contributors
    - Code: github.com/vhive-serverless/vHive
- Simple device simulation in Simics for the "Fundamentals of Full-Platform Simulation" course at MIPT
    - Device model development using Simics’ DML, creation of Linux drivers for the device
    - Code: github.com/leokondrashov/simics_device

Certificates and Awards

- IELTS Academic, 7.5 total score (10.2022)
- Abramov and Frolov scholarship at MIPT (09.2019 - 05.2021)
- 2nd place in the team olympiad NTI in the area of Wireless Technologies (04.2020)
- Certificate with honours for Technotrack course “Industrial programming in C” (01.2019)